

return, variance drain, growth-optimal investing, volatility pumping, and the Kelly criterion or Kelly rule, named after John Kelly (1956), an engineer who worked at Bell Labs.³⁶ The term “diversification return” was introduced by Booth and Fama (1992) and is probably the best-known term in finance, whereas the Kelly rule and volatility pumping are better known in mathematics. I prefer not to use Booth and Fama’s terminology because there is a difference between diversification in a single period and rebalancing, which earns a premium over time. Diversification gets you a benefit in one period, but this diversification benefit dies out if you do not rebalance.³⁷ The rebalancing premium only exists for a long-horizon investor, and he can collect it by rebalancing to constant weights every period. I use the term “rebalancing premium” to emphasize that the premium comes from rebalancing, not from diversification.

5.1. REBALANCING BEATS BUY-AND-HOLD OVER THE LONG RUN

Suppose that the price of each underlying asset is *stationary*; that is, the price of each asset tends to hover around a fixed range and never goes off to infinity. Holding 100% positions in each asset never gives you increasing wealth. But, a rebalanced portfolio does give you wealth that increases over the long run to infinity (wealth increases *exponentially fast*).³⁸ Furthermore, by rebalancing to a fixed constant weight each period, an investor can generate wealth that increases over time, and *any* such rebalancing strategy will eventually beat the best buy-and-hold portfolio. This seems like magic: Erb and Harvey (2006) call it “turning water into wine” and Evstigneev and Schenk-Hoppé (2002) call it going from “rags to riches.”

Mathematically, this is not quite as impressive as Jesus’ first miracle at the wedding at Cana. It arises as a consequence of compounding. We can see this in equation (4.7), where long-term wealth is a product of arithmetic returns, $(1 + r_t)(1 + r_{t+1})(1 + r_{t+2}) \dots$, rather than a sum of arithmetic returns, $(1 + r_t)(1 + r_{t+1})(1 + r_{t+2}) \dots \neq 1 + r_t + r_{t+1} + r_{t+2} + \dots$. The compounding of products gives rise to many nonlinearities over time, which are called *Jensen’s terms*,³⁹ and the effect of the nonlinear terms increases over time. The entire rebalancing premium is due to Jensen’s terms, and in fact the whole diversification return and Kelly rule literature can be viewed as a paean to *Jensen’s inequality*.

³⁶ For growth-optimal investing, see Latané (1959) and Messmore (1995) for the variance drain terminology. Luenberger (1997) introduced the term “volatility pumping.” A nice collection of papers in the literature is MacLean, Thorp, and Ziemba (2011).

³⁷ See also Willenbrock (2011) who differentiates between diversification as being necessary to give you different weights over one period but not sufficient to earn the rebalancing premium over multiple periods.

³⁸ This is true also for modest transaction costs, as shown by Dempster, Evstigneev, and Schenk-Hoppé (2009).

³⁹ Named after the Danish mathematician Johan Jensen.